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https://www.tensorflow.org/api_docs/python/tf/IndexSlices

A sparse representation of a set of tensor slices at given indices.

Function Signatures

```
tf.IndexedSlices( values, indices, dense_shape=None )
```

Code Examples

```
tf.IndexedSlices(  
    values, indices, dense_shape=None  
)
```

```
tf.IndexedSlices(  
    values, indices, dense_shape=None  
)
```

```
dense[slices.indices[i], :, :, :, ...] = slices.values[i, :, :, :, ...]
```

```
dense[slices.indices[i], :, :, :, ...] = slices.values[i, :, :, :, ...]
```

```
v = tf.Variable([[0.,1, 2], [2, 3, 4], [4, 5, 6], [6, 7, 8]])  
with tf.GradientTape() as tape:  
  r = tf.gather(v, [1,3])  
index_slices = tape.gradient(r,v)  
index_slices  
<...IndexedSlices object ...>  
index_slices.indices.numpy()  
array([1, 3], dtype=int32)  
index_slices.values.numpy()  
array([[1., 1., 1.],  
       [1., 1., 1.]], dtype=float32)
```

```
v = tf.Variable([[0.,1, 2], [2, 3, 4], [4, 5, 6], [6, 7, 8]])
```

```
with tf.GradientTape() as tape:
```

```
  r = tf.gather(v, [1,3])
```

Documentation

A sparse representation of a set of tensor slices at given indices.

View aliases

Compat aliases for migration

See

Migration guide for

more details.

`tf.compat.v1.IndexedSlices`

Used in the notebooks

- Client-efficient large-model federated learning via ``federated_select`` and sparse aggregation

This class is a simple wrapper for a pair of Tensor objects:

- `values`: A Tensor of any dtype with shape `[D0, D1, ..., Dn]`.
- `indices`: A 1-D integer Tensor with shape `[D0]`.

An `IndexedSlices` is typically used to represent a subset of a larger tensor dense of shape `[LARGE0, D1, .. , DN]` where `LARGE0 >> D0`.

The values in `indices` are the indices in the first dimension of the slices that have been extracted from the larger tensor.

The dense tensor dense represented by an `IndexedSlices` slices has

The `IndexedSlices` class is used principally in the definition of gradients for operations that have sparse gradients (e.g. `tf.gather`).

Contrast this representation with

`tf.sparse.SparseTensor`,

which uses multi-dimensional indices and scalar values.

Attributes

[## Methods](#)

consumers

[View source](#)

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